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Efficient inference for conditional longitudinal modified treatment policies

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Summary

Longitudinal modified treatment policies (LMTPs) are an increasingly common approach for the estimation and inference of causal effects corresponding to changes in longitudinal exposures or treatment. In many problems, scientific interest lies not only in overall effects but also subgroup-specific or conditional effects. However, little work has been done to address efficiently estimating the effects of LMTPs on many subgroups simultaneously. We present a novel targeted learning approach for the efficient estimation of subgroup-specific LMTP effects, where subgroups are defined by characteristics of observations at baseline, which provides multiply robust and semiparametric efficient estimators. Our procedure for estimating subgroup effects of LMTPs allows for the construction of simultaneous confidence intervals across subgroups. For handling continuous covariates defined at baseline, we develop a DR-learner inspired approach for estimating conditional LMTPs consisting of forming an efficient influence function based pseudo-outcome regressed on baseline variables. As a case study, we apply our method to the question of how deforestation in Madagascar during the 2010-2022 timeframe may have

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affected malaria incidence rates at 2,773 clinics where effect heterogeneity was expected by ecological zone.

Keywords: Causal inference, Longitudinal Modified Treatment Policies, Treatment Effect Heterogeneity, Efficient Inference, Subgroup Effects

1. INTRODUCTION

Longitudinal studies in medicine, epidemiology, and the environmental sciences often seek to evaluate interventions on exposures or treatments that evolve over time. In such settings, treatment decisions may depend on a subject's previous treatment and covariate history, and time-varying covariates may themselves be affected by earlier treatment. These features rule out standard regression adjustment and motivate the use of causal estimands defined through longitudinal interventions.

Longitudinal modified treatment policies (LMTPs) provide one useful class of such interventions. Rather than setting treatment to one fixed value, an LMTP modifies the treatment that a subject would otherwise have received according to a user-specified policy map. This makes LMTPs attractive in settings where deterministic static interventions are implausible or scientifically less relevant. For example, in studies of environmental exposures, one may be interested in shifting an exposure downward for only units above a threshold; and in health studies, modifying a dose or treatment intensity relative to the value actually received. LMTPs therefore provide a flexible way to encode realistic longitudinal interventions while retaining identification under standard sequential causal assumptions.

Most methodological work on LMTPs has focused on marginal causal parameters, such as the population mean outcome under a single policy or the contrast between two policies. However, scientific questions about policy effects are often inherently conditional. Investigators may ask whether an intervention has larger effects in certain ecological zones, clinical subgroups, baseline risk strata, or other baseline-defined populations. More generally, one may wish to estimate a conditional effect curve describing how the causal effect of a longitudinal policy varies with baseline covariates. Restricting attention to only marginal LMTP effects can obscure such heterogeneity.

This paper studies estimation and inference for conditional effects of LMTPs. Let V denote baseline covariates with respect to which effect heterogeneity is of interest. For an LMTP d, we define the conditional LMTP mean

$$\theta_{\mathrm{d}}(v) = \mathbb{E}\{Y(\mathrm{d}) \mid V = v\},$$

and, for two policies d_1 and d_0, the conditional LMTP contrast

$$\tau(v) = \theta_{\mathrm{d}_1}(v) - \theta_{\mathrm{d}_0}(v).$$

When V is discrete, this target corresponds to a vector of subgroup-specific policy effects. When V is continuous or high-dimensional, it corresponds to a conditional effect function.

We will first present a formal identification argument for conditional LMTP means and contrasts using the observed-data likelihood factorization, the post-policy law, and the longitudinal g-formula. This yields inverse-probability and iterated-regression representations of the conditional LMTP effect. Second, for finite baseline subgroups, we develop a targeted learning estimator for the vector of subgroup LMTP means or contrasts. This estimator solves the efficient influence-function estimating equations for all subgroups jointly and supports Wald-type and simultaneous inference. Third, for general condi-

tioning covariates V, we develop a DR-Learner for conditional LMTP effects. The DR-Learner estimates uncentered influence-function pseudo-outcomes for marginal LMTP parameters and then regresses these pseudo-outcomes on V. We establish conditions under which this estimator is asymptotically equivalent to the benchmark second-stage regression that would be obtained if the nuisance functions were known.

The remainder of the paper is organized as follows. Section 2 defines the observed data structure, LMTPs, conditional target parameters, identification assumptions, identification results and derives the pseudo-outcome representation used by the DR-Learner. Section 3 presents the finite-subgroup targeted minimum loss-based estimator and the conditional LMTP DR-Learner. Section 4 gives the large-sample properties of the proposed estimators. Section **??** presents simulation studies, and Section 6 applies the methods to estimate heterogeneous effects of deforestation on malaria incidence in Madagascar. Section 7 offers discussion of potential applications for conditional LMTP estimation and future directions for conditional effect estimation in longitudinal causal inference.

2. Setup and Target Parameters

2.1. Observed data and likelihood factorization

Let

$$O = (L_0, A_0, L_1, A_1, \ldots, L_\tau, A_\tau, Y)$$

denote the observed data for a randomly sampled unit. Here L_0 denotes baseline covariates, A_t denotes the treatment or exposure at time $t = 0, \ldots, \tau$, and L_{t+1} denotes covariates measured after treatment A_t, with $L_{\tau+1} \equiv Y$. Let

$$\overline{L}_t = (L_0, \ldots, L_t), \qquad \overline{A}_t = (A_0, \ldots, A_t),$$

and define the treatment history available immediately before treatment time t by

$$H_t = (\overline{L}_t, \overline{A}_{t-1}),$$

with the convention that $H_0 = L_0$. We assume throughout that $O_1, \ldots, O_n$ are independent and identically distributed according to an unknown law P.

Assume that P is dominated by a product measure and has density

$$p(o) = q_0(l_0) \prod_{t=0}^{\tau} g_t(a_t \mid h_t) q_{t+1}(l_{t+1} \mid h_t, a_t),$$

where

$$g_t(a_t \mid h_t) = p(A_t = a_t \mid H_t = h_t)$$

denotes the conditional density or probability mass function of the treatment at time t, and

$$q_{t+1}(l_{t+1} \mid h_t, a_t) = p(L_{t+1} = l_{t+1} \mid H_t = h_t, A_t = a_t)$$

denotes the conditional law of the next covariate, or of the final outcome when $t = \tau$. We write

$$q = (q_0, q_1, \ldots, q_{\tau+1}), \qquad g = (g_0, \ldots, g_\tau),$$

so that $p = qg$. This factorization separates the conditional treatment process g from the covariate and outcome transition laws q. The causal parameters considered below are

identified by replacing the observed treatment process g with a post-policy treatment process while leaving q unchanged.

2.2. Longitudinal modified treatment policies

A longitudinal modified treatment policy is a sequence

$$\mathrm{d} = (d_0, \ldots, d_\tau)$$

of known maps. At time t, the map d_t takes an observed treatment value and history (a_t, h_t) and returns the treatment value that would be assigned under the policy,

$$a_t^{\mathrm{d}} = d_t(a_t, h_t).$$

The corresponding random variable is

$$A_t^{\mathrm{d}} = d_t(A_t, H_t).$$

For example, with a continuous exposure, d_t might shift exposure downward by a fixed amount among units above a threshold. With a discrete exposure, d_t might map some observed treatment levels to modified levels while leaving other levels unchanged.

In the longitudinal setting, the policy is evaluated recursively. Earlier modified treatments affect later histories, and later modified treatments are therefore functions of histories that include previously modified treatment values. Let

$$H_t^{\mathrm{d}} = (\overline{L}_t, A_0^{\mathrm{d}}, \ldots, A_{t-1}^{\mathrm{d}})$$

denote the history through time t under the policy. Then the policy-modified treatment process is generated recursively by

$$A_0^{\mathrm{d}} = d_0(A_0, L_0),$$

and, for $t = 1, \ldots, \tau$,

$$A_t^{\mathrm{d}} = d_t\{A_t(\overline{A}_{t-1}^{\mathrm{d}}), H_t^{\mathrm{d}}\},$$

where $A_t(\overline{A}_{t-1}^{\mathrm{d}})$ denotes the treatment value that would have been observed at time t had treatment through time $t-1$ been modified according to d.

Let

$$g_t^{\mathrm{d}}(a_t \mid h_t)$$

denote the conditional law of A_t^{d} given $H_t = h_t$. For discrete treatments, g_t^{d} is the push-forward of the observed treatment law g_t under the map d_t. For continuous treatments, we assume that $d_t(\cdot, h_t)$ is piecewise smooth and invertible on a finite partition of the treatment space, so that g_t^{d} is well-defined by the change-of-variables formula.

The post-policy distribution P^{d} is the law with density

$$p^{\mathrm{d}}(o) = q_0(l_0) \prod_{t=0}^{\tau} g_t^{\mathrm{d}}(a_t \mid h_t) q_{t+1}(l_{t+1} \mid h_t, a_t).$$

Thus the intervention changes the conditional treatment law at each time point but leaves the conditional covariate and outcome transition laws unchanged. This is the usual longitudinal g-formula representation specialized to modified treatment policies.

2.3. Conditional LMTP means and contrasts

Let $V \subseteq L_0$ denote baseline covariates with respect to which effect heterogeneity is of interest. The covariates V may be a discrete subgroup indicator, a vector of baseline features, or a lower-dimensional summary of baseline risk. Because V is measured before treatment, its distribution is not changed by the intervention.

For a single LMTP d, define the marginal LMTP mean

$$\psi_{\mathrm{d}} = \mathbb{E}_{P^{\mathrm{d}}}(Y) = \mathbb{E}\{Y(\mathrm{d})\}.$$

The corresponding conditional LMTP mean is

$$\theta_{\mathrm{d}}(v) = \mathbb{E}_{P^{\mathrm{d}}}(Y \mid V = v) = \mathbb{E}\{Y(\mathrm{d}) \mid V = v\}.$$

For two policies d_1 and d_0, define the conditional LMTP contrast

$$\tau(v) = \theta_{\mathrm{d}_1}(v) - \theta_{\mathrm{d}_0}(v) = \mathbb{E}\{Y^{\mathrm{d}_1} - Y^{\mathrm{d}_0} \mid V = v\}.$$

When V is discrete with support $\mathcal{V}$, the target is the finite collection

$$\{\theta_{\mathrm{d}}(v) : v \in \mathcal{V}\}$$

or, for a contrast of two policies,

$$\{\tau(v) : v \in \mathcal{V}\}.$$

In this case, the target is a finite-dimensional parameter, and standard semiparametric efficiency theory can be used to construct asymptotically linear estimators and Wald-type confidence intervals.

When V is continuous, the target $v \mapsto \theta_{\mathrm{d}}(v)$ or $v \mapsto \tau(v)$ is an infinite-dimensional conditional mean function. In this case, the conditional effect itself is not generally a pathwise differentiable parameter at a point v without additional smoothing or projection. We therefore distinguish between finite-dimensional subgroup inference and flexible estimation of conditional effect functions.

2.4. Identification assumptions

Identification of the conditional LMTP mean relies on a longitudinal causal model, a support condition for the modified treatment policy, and regularity conditions ensuring that the post-policy treatment density is well-defined.

We first introduce a nonparametric structural equation model to state the sequential randomization condition precisely. Suppose the observed data are generated by structural equations

$$L_0 = f_{L_0}(U_{L_0}),$$

and, for $t = 0, \ldots, \tau$,

$$A_t = f_{A_t}(H_t, U_{A_t}), \qquad L_{t+1} = f_{L_{t+1}}(H_t, A_t, U_{L_{t+1}}),$$

where $L_{\tau+1} = Y$. For $t = 0, \ldots, \tau$, define

$$\underline{U}_{L,t+1} = (U_{L_{t+1}}, \ldots, U_{L_{\tau+1}}), \qquad \underline{U}_{A,t+1} = (U_{A_{t+1}}, \ldots, U_{A_\tau}).$$

The exogenous variables encode sources of variation not included in the observed history.

Assumption A1 *(Consistency).* The observed data are generated by the structural equations above. For any treatment history $\overline{a}_\tau$, the counterfactual outcome $Y(\overline{a}_\tau)$ is obtained by replacing the treatment equations with $A_t = a_t$, $t = 0, \ldots, \tau$, while leaving

the covariate and outcome equations unchanged. In particular, for the observed treatment history,

$$Y = Y(\overline{A}_\tau).$$

Assumption A2 *(Strong sequential randomization).* For each $t = 0, \ldots, \tau$,

$$U_{A_t} \perp\!\!\!\perp \left(\underline{U}_{L,t+1}, \underline{U}_{A,t+1}\right) \mid H_t.$$

Equivalently, conditional on the observed history H_t, the unobserved determinants of treatment at time t are independent of the unobserved determinants of future covariates, future treatments, and the final outcome.

Remark 1. Assumption A2 is the NPSEM version of no unmeasured time-varying confounding. It implies the more familiar sequential exchangeability statement that, conditional on H_t, the treatment actually received at time t is independent of future counterfactual variables under the treatment regimes considered in the analysis.

Assumption A3 *(LMTP support condition).* For each $t = 0, \ldots, \tau$, if

$$(a_t, h_t) \in \operatorname{supp}(A_t, H_t),$$

then

$$\{d_t(a_t, h_t), h_t\} \in \operatorname{supp}(A_t, H_t).$$

Equivalently, for P-almost every history h_t, the modified treatment value $d_t(a_t, h_t)$ lies in the conditional support of $A_t \mid H_t = h_t$ whenever a_t lies in that support.

Remark 2. Assumption A3 is the LMTP analogue of positivity. It is weaker and more local than the usual static-treatment positivity requirement. It does not require every treatment level to be possible at every history. Instead, it requires only that the treatment value assigned by the modified policy be observable, in principle, among units with the same history.

Assumption A4 *(Regularity of the policy map).* For each $t = 0, \ldots, \tau$, the map $d_t(\cdot, h_t)$ is known. If A_t is continuous, then $d_t(\cdot, h_t)$ is piecewise smooth and invertible on a finite partition of the treatment space for P-almost every h_t. On each piece, the inverse map $d_t^{-1}(\cdot, h_t)$ exists and has derivative bounded away from zero and infinity on the relevant support. If A_t is discrete, the pushforward law induced by d_t is well-defined.

Under Assumption A4, the post-policy treatment density is defined as the law of $A_t^{\mathrm{d}} = d_t(A_t, H_t)$ conditional on H_t. For continuous treatments, this density is obtained by the change-of-variables formula. In the one-to-one case,

$$g_t^{\mathrm{d}}(a_t \mid h_t) = g_t\{d_t^{-1}(a_t, h_t) \mid h_t\} \left| \frac{\partial}{\partial a_t} d_t^{-1}(a_t, h_t) \right|.$$

For piecewise invertible maps, the density is the sum of the corresponding piecewise inverse contributions. For discrete treatments,

$$g_t^{\mathrm{d}}(a_t \mid h_t) = \sum_{a_t' :\, d_t(a_t', h_t) = a_t} g_t(a_t' \mid h_t).$$

Assumption A5 *(Baseline conditioning event).* For discrete V, assume

$$p_v = \mathbb{P}(V = v) > 0$$

for every subgroup value v under consideration. For continuous V, $\theta_{\mathrm{d}}(v)$ is interpreted as a version of $\mathbb{E}_{P^{\mathrm{d}}}(Y \mid V = v)$, defined for v in the support of V.

Assumptions A1 and A2 give the causal interpretation of the post-policy law. Assumption A3 ensures that the policy does not assign treatment values outside the observed conditional support. Assumption A4 ensures that the post-policy treatment law g_t^{d} and density ratios used for estimation are well-defined. Assumption A5 ensures that the conditional target is defined for the subgroup or covariate values of interest.

2.5. Iterated conditional mean representation

Define the terminal outcome regression

$$m_\tau^{\mathrm{d}}(a_\tau, h_\tau) = \mathbb{E}_P\left[Y \mid A_\tau = a_\tau, H_\tau = h_\tau\right].$$

For $t = \tau - 1, \ldots, 0$, define the sequential regressions recursively by

$$m_t^{\mathrm{d}}(a_t, h_t) = \mathbb{E}_P\left[m_{t+1}^{\mathrm{d}}\{A_{t+1}^{\mathrm{d}}, H_{t+1}\} \,\middle|\, A_t = a_t, H_t = h_t\right].$$

Here

$$A_{t+1}^{\mathrm{d}} = d_{t+1}(A_{t+1}, H_{t+1})$$

is the treatment value assigned by the modified treatment policy at time $t+1$, evaluated using the observed history H_{t+1}. This recursion is the iterated conditional mean version of the longitudinal g-formula.

With this notation, the marginal LMTP mean is identified as

$$\psi_{\mathrm{d}} = \mathbb{E}_P\left[m_0^{\mathrm{d}}(A_0^{\mathrm{d}}, L_0)\right],$$

and the conditional LMTP mean is identified as

$$\theta_{\mathrm{d}}(v) = \mathbb{E}_P\left[m_0^{\mathrm{d}}(A_0^{\mathrm{d}}, L_0) \mid V = v\right].$$

Theorem 1 (Iterated conditional mean identification). *Under the assumptions A1–A5,*

$$\theta_{\mathrm{d}}(v) = \mathbb{E}_P\left[m_0^{\mathrm{d}}(A_0^{\mathrm{d}}, L_0) \mid V = v\right]. \tag{1}$$

For two policies d_1 *and* d_0,

$$\tau(v) = \mathbb{E}_P\left[m_0^{\mathrm{d}_1}(A_0^{\mathrm{d}_1}, L_0) - m_0^{\mathrm{d}_0}(A_0^{\mathrm{d}_0}, L_0) \mid V = v\right].$$

Proof. The result follows by repeated application of the law of iterated expectations. At the final time point,

$$m_\tau^{\mathrm{d}}(a_\tau, h_\tau) = \mathbb{E}_P(Y \mid A_\tau = a_\tau, H_\tau = h_\tau)$$

is the conditional mean of the outcome given the final treatment and history. At time $\tau - 1$, the regression

$$m_{\tau-1}^{\mathrm{d}}(a_{\tau-1}, h_{\tau-1}) = \mathbb{E}_P\left[m_\tau^{\mathrm{d}}(A_\tau^{\mathrm{d}}, H_\tau) \mid A_{\tau-1} = a_{\tau-1}, H_{\tau-1} = h_{\tau-1}\right]$$

integrates the final outcome regression over the conditional law of the next covariate and treatment, with the next treatment evaluated under the modified policy. Repeating this argument backward through $t = \tau - 2, \ldots, 0$ gives the g-formula mean under P^{d}. Conditioning on $V = v$ gives 1. The contrast result follows by applying the same argument to each policy and taking the difference. □

2.6. *Uncentered influence-function pseudo-outcomes*

The preceding representations identify the conditional LMTP mean. For estimation, it is useful to construct a pseudo-outcome whose conditional expectation is the target function $v \mapsto \theta_{\mathrm{d}}(v)$. This pseudo-outcome is obtained from the efficient influence function for the marginal LMTP mean by adding back the marginal mean.

Let

$$\eta^{\mathrm{d}} = \left(m_0^{\mathrm{d}}, \ldots, m_\tau^{\mathrm{d}}, r_0^{\mathrm{d}}, \ldots, r_\tau^{\mathrm{d}}\right)$$

denote the nuisance functions for policy d. Define

$$\omega_s^{\mathrm{d}}(O) = \prod_{k=0}^{s} r_k^{\mathrm{d}}(A_k, H_k), \qquad s = 0, \ldots, \tau.$$

For $s = 0, \ldots, \tau - 1$, define the sequential residual

$$\Delta_s^{\mathrm{d}}(O; \eta^{\mathrm{d}}) = m_{s+1}^{\mathrm{d}}\{A_{s+1}^{\mathrm{d}}, H_{s+1}\} - m_s^{\mathrm{d}}(A_s, H_s),$$

and define the terminal residual

$$\Delta_\tau^{\mathrm{d}}(O; \eta^{\mathrm{d}}) = Y - m_\tau^{\mathrm{d}}(A_\tau, H_\tau).$$

Thus each residual compares the next-stage pseudo-outcome to its conditional mean given the current treatment and history.

The uncentered influence-function pseudo-outcome is

$$\varphi_{\mathrm{d}}(O; \eta^{\mathrm{d}}) = m_0^{\mathrm{d}}(A_0^{\mathrm{d}}, L_0) + \sum_{s=0}^{\tau} \omega_s^{\mathrm{d}}(O) \Delta_s^{\mathrm{d}}(O; \eta^{\mathrm{d}}).$$

Equivalently,

$$\varphi_{\mathrm{d}}(O; \eta^{\mathrm{d}}) = m_0^{\mathrm{d}}(A_0^{\mathrm{d}}, L_0) + \sum_{s=0}^{\tau-1} \left\{ \prod_{k=0}^{s} r_k^{\mathrm{d}}(A_k, H_k) \right\} \left[m_{s+1}^{\mathrm{d}}\{A_{s+1}^{\mathrm{d}}, H_{s+1}\} - m_s^{\mathrm{d}}(A_s, H_s) \right] + \left\{ \prod_{k=0}^{\tau} r_k^{\mathrm{d}}(A_k, H_k) \right\} \left[Y - m_\tau^{\mathrm{d}}(A_\tau, H_\tau) \right].$$

The product in this expression is the product of density ratios up to time s; the terms in square brackets are sequential regression residuals. Thus the EIF is a sum of weighted residuals.

At the true nuisance functions, the centered efficient influence function for the marginal LMTP mean ψ_{d} is

$$D_{\mathrm{d}}(O) = \varphi_{\mathrm{d}}(O; \eta^{\mathrm{d}}) - \psi_{\mathrm{d}}.$$

Equivalently,

$$\varphi_{\mathrm{d}}(O; \eta^{\mathrm{d}}) = \psi_{\mathrm{d}} + D_{\mathrm{d}}(O).$$

Therefore

$$\mathbb{E}_P\{\varphi_{\mathrm{d}}(O; \eta^{\mathrm{d}})\} = \psi_{\mathrm{d}}.$$

The key property for conditional effect estimation is that, since $V \subseteq L_0$, the conditional mean of the uncentered pseudo-outcome is the conditional LMTP mean.

THEOREM 2 (PSEUDO-OUTCOME REPRESENTATION OF THE CONDITIONAL LMTP MEAN). *Under Assumptions A1–A5,*

$$\mathbb{E}_P\left[\varphi_{\mathrm{d}}(O;\eta^{\mathrm{d}}) \mid V = v\right] = \theta_{\mathrm{d}}(v).$$

Therefore, for two policies d_1 *and* d_0*, if*

$$\varphi_\tau(O;\eta) = \varphi_{\mathrm{d}_1}(O;\eta^{\mathrm{d}_1}) - \varphi_{\mathrm{d}_0}(O;\eta^{\mathrm{d}_0}),$$

then

$$\mathbb{E}_P\left[\varphi_\tau(O;\eta) \mid V = v\right] = \tau(v).$$

Proof. By Theorem 1,

$$\mathbb{E}_P\left[m_0^{\mathrm{d}}(A_0^{\mathrm{d}}, L_0) \mid V = v\right] = \theta_{\mathrm{d}}(v).$$

It remains to show that the weighted residual terms have conditional mean zero given V.

Fix $s \in \{0, \ldots, \tau - 1\}$. Since $\omega_s^{\mathrm{d}}(O)$ is measurable with respect to (A_s, H_s), and since $V \subseteq L_0 \subseteq H_s$, iterated expectation gives

$$\mathbb{E}_P\left[\omega_s^{\mathrm{d}}(O)\Delta_s^{\mathrm{d}}(O;\eta^{\mathrm{d}}) \mid V\right] = \mathbb{E}_P\left[\omega_s^{\mathrm{d}}(O)\mathbb{E}_P\left\{\Delta_s^{\mathrm{d}}(O;\eta^{\mathrm{d}}) \mid A_s, H_s\right\} \mid V\right].$$

By the definition of m_s^{d},

$$\mathbb{E}_P\left[m_{s+1}^{\mathrm{d}}\{A_{s+1}^{\mathrm{d}}, H_{s+1}\} \mid A_s, H_s\right] = m_s^{\mathrm{d}}(A_s, H_s),$$

so

$$\mathbb{E}_P\left\{\Delta_s^{\mathrm{d}}(O;\eta^{\mathrm{d}}) \mid A_s, H_s\right\} = 0.$$

Therefore

$$\mathbb{E}_P\left[\omega_s^{\mathrm{d}}(O)\Delta_s^{\mathrm{d}}(O;\eta^{\mathrm{d}}) \mid V\right] = 0.$$

For the terminal term,

$$\mathbb{E}_P\left[Y - m_\tau^{\mathrm{d}}(A_\tau, H_\tau) \mid A_\tau, H_\tau\right] = 0$$

by the definition of m_τ^{d}. The same iterated-expectation argument gives

$$\mathbb{E}_P\left[\omega_\tau^{\mathrm{d}}(O)\Delta_\tau^{\mathrm{d}}(O;\eta^{\mathrm{d}}) \mid V\right] = 0.$$

Thus all weighted residual terms in $\varphi_{\mathrm{d}}(O;\eta^{\mathrm{d}})$ have conditional mean zero given V. Hence

$$\mathbb{E}_P\left[\varphi_{\mathrm{d}}(O;\eta^{\mathrm{d}}) \mid V = v\right] = \mathbb{E}_P\left[m_0^{\mathrm{d}}(A_0^{\mathrm{d}}, L_0) \mid V = v\right] = \theta_{\mathrm{d}}(v).$$

The contrast result follows by linearity of conditional expectation. □

Remark 3 *(Role of the pseudo-outcome).* The pseudo-outcome $\varphi_{\mathrm{d}}(O;\eta^{\mathrm{d}})$ is not a new causal parameter. It is a representation of the same identified LMTP mean. Its role is statistical: it converts estimation of the conditional LMTP mean into a regression problem. For a general conditioning variable V, the DR-Learner estimates

$$v \mapsto \mathbb{E}\left[\varphi_{\mathrm{d}}(O;\eta^{\mathrm{d}}) \mid V = v\right]$$

by replacing the unknown nuisance functions with estimates and then regressing the resulting estimated pseudo-outcomes on V.

3. ESTIMATION

We now describe estimators for the conditional LMTP parameters introduced in Section 2. We consider two complementary estimation strategies. First, when V is discrete with finite support $\mathcal{V}$, we construct a targeted minimum loss-based estimator of the finite-dimensional vector

$$\{\theta_{\mathbb{d}}(v) : v \in \mathcal{V}\}.$$

This estimator targets the efficient influence-function estimating equations for all subgroups jointly. Second, when V is continuous, high-dimensional, or when the goal is to estimate a smooth conditional effect function, we construct a DR-Learner by regressing estimated uncentered influence-function pseudo-outcomes on V.

Throughout this section, let

$$\eta^{\mathbb{d}} = \left(m_0^{\mathbb{d}}, \ldots, m_\tau^{\mathbb{d}}, r_0^{\mathbb{d}}, \ldots, r_\tau^{\mathbb{d}}\right)$$

denote the nuisance functions for policy $\mathbb{d}$. For two policies $\mathbb{d}_1$ and $\mathbb{d}_0$, all steps below may be applied separately to each policy and then contrasted.

3.1. Cross-fitting and nuisance estimation

Flexible estimation of the nuisance functions is essential in longitudinal settings, where the treatment mechanism and sequential outcome regressions may depend on high-dimensional histories. We use cross-fitting to separate nuisance estimation from evaluation of the estimating equations. This allows the nuisance functions to be estimated with flexible regression or machine-learning methods without requiring empirical process restrictions such as Donsker conditions.

Let

$$\mathcal{I}_1, \ldots, \mathcal{I}_J$$

be a partition of $\{1, \ldots, n\}$ into J folds. For each fold j, let $\mathcal{I}_{-j}$ denote its complement. Nuisance functions are fit using only the training sample $\mathcal{I}_{-j}$, and predictions are evaluated only on the validation sample $\mathcal{I}_j$. For observation i, let $j(i)$ denote the fold containing i.

For each policy $\mathbb{d}$, each fold j, and each time point $t = 0, \ldots, \tau$, estimate the density ratio

$$r_t^{\mathbb{d}}(a_t, h_t) = \frac{g_t^{\mathbb{d}}(a_t \mid h_t)}{g_t(a_t \mid h_t)}$$

using the training sample $\mathcal{I}_{-j}$. Denote the resulting cross-fitted prediction for observation i by

$$\widehat{r}_{t,i}^{\mathbb{d}} = \widehat{r}_{t,j(i)}^{\mathbb{d}}(A_{t,i}, H_{t,i}).$$

The estimated cumulative density ratio through time s is

$$\widehat{\omega}_{s,i}^{\mathbb{d}} = \prod_{t=0}^{s} \widehat{r}_{t,i}^{\mathbb{d}}.$$

The sequential outcome regressions are also estimated recursively using cross-fitting. Starting at the final time point, estimate

$$m_\tau^{\mathrm{d}}(a_\tau, h_\tau) = \mathbb{E}(Y \mid A_\tau = a_\tau, H_\tau = h_\tau)$$

on $\mathcal{I}_{-j}$, and evaluate it on $\mathcal{I}_j$ at both the observed and policy-modified treatment values as required. For $t = \tau - 1, \ldots, 0$, estimate

$$m_t^{\mathrm{d}}(a_t, h_t) = \mathbb{E}\left[m_{t+1}^{\mathrm{d}}\{A_{t+1}^{\mathrm{d}}, H_{t+1}\} \mid A_t = a_t, H_t = h_t\right].$$

In practice, the regression outcome at time t is the previously estimated next-stage pseudo-outcome

$$\widehat{m}_{t+1,j}^{\mathrm{d}}\{A_{t+1}^{\mathrm{d}}, H_{t+1}\},$$

constructed using the training sample. The resulting cross-fitted predictions for observation i are denoted by

$$\widehat{m}_{t,i}^{\mathrm{d}}(A_{t,i}, H_{t,i}) \qquad \text{and} \qquad \widehat{m}_{t,i}^{\mathrm{d}}(A_{t,i}^{\mathrm{d}}, H_{t,i}).$$

Given these nuisance estimates, define the cross-fitted estimated residuals

$$\widehat{\Delta}_{s,i}^{\mathrm{d}} = \widehat{m}_{s+1,i}^{\mathrm{d}}\{A_{s+1,i}^{\mathrm{d}}, H_{s+1,i}\} - \widehat{m}_{s,i}^{\mathrm{d}}(A_{s,i}, H_{s,i}), \qquad s = 0, \ldots, \tau - 1,$$

and

$$\widehat{\Delta}_{\tau,i}^{\mathrm{d}} = Y_i - \widehat{m}_{\tau,i}^{\mathrm{d}}(A_{\tau,i}, H_{\tau,i}).$$

The estimated uncentered influence-function pseudo-outcome is then

$$\widehat{\varphi}_{\mathrm{d},i} = \widehat{m}_{0,i}^{\mathrm{d}}(A_{0,i}^{\mathrm{d}}, L_{0,i}) + \sum_{s=0}^{\tau} \widehat{\omega}_{s,i}^{\mathrm{d}} \widehat{\Delta}_{s,i}^{\mathrm{d}}.$$

For a contrast of two policies, define

$$\widehat{\varphi}_{\tau,i} = \widehat{\varphi}_{\mathrm{d}_1,i} - \widehat{\varphi}_{\mathrm{d}_0,i}.$$

Remark 4. The notation $\widehat{\varphi}_{\tau,i}$ denotes the estimated pseudo-outcome for the conditional contrast $\tau(v)$, not a time-indexed quantity. When there is risk of confusion, we write $\widehat{\varphi}_{\mathrm{d}_1-\mathrm{d}_0,i}$.

3.2. Targeted estimation for finite baseline subgroups

Suppose V is discrete with finite support

$$\mathcal{V} = \{v_1, \ldots, v_K\},$$

and let

$$p_v = \mathbb{P}(V = v).$$

The conditional LMTP mean for subgroup v can be written as

$$\theta_{\mathrm{d}}(v) = \frac{\mathbb{E}\left[\mathbb{1}(V = v) m_0^{\mathrm{d}}(A_0^{\mathrm{d}}, L_0)\right]}{p_v}.$$

The efficient influence function for $\theta_{\mathrm{d}}(v)$ is

$$D_{\mathrm{d},v}(O) = \frac{\mathbb{1}(V = v)}{p_v}\left\{\varphi_{\mathrm{d}}(O; \eta^{\mathrm{d}}) - \theta_{\mathrm{d}}(v)\right\}.$$

Equivalently,

$$D_{\mathrm{d},v}(O) = \frac{\mathbb{1}(V=v)}{p_v}\left[m_0^{\mathrm{d}}(A_0^{\mathrm{d}}, L_0) - \theta_{\mathrm{d}}(v) + \sum_{s=0}^{\tau}\omega_s^{\mathrm{d}}(O)\Delta_s^{\mathrm{d}}(O;\eta^{\mathrm{d}})\right].$$

For a contrast

$$\tau(v) = \theta_{\mathrm{d}_1}(v) - \theta_{\mathrm{d}_0}(v),$$

the efficient influence function is

$$D_{\tau,v}(O) = D_{\mathrm{d}_1,v}(O) - D_{\mathrm{d}_0,v}(O).$$

The subgroup TMLE targets the sequential outcome regressions so that the empirical mean of the estimated efficient influence function is approximately zero for each $v \in \mathcal{V}$. Let

$$\widehat{p}_v = \mathbb{P}_n\{\mathbb{1}(V=v)\}.$$

For $s = 0, \ldots, \tau$, define the subgroup-specific clever covariate

$$\widehat{H}_{s,v}^{\mathrm{d}}(O_i) = \frac{\mathbb{1}(V_i=v)}{\widehat{p}_v}\widehat{\omega}_{s,i}^{\mathrm{d}}.$$

Collecting these across $v \in \mathcal{V}$ gives the vector clever covariate

$$\widehat{H}_s^{\mathrm{d}}(O_i) = \left(\widehat{H}_{s,v_1}^{\mathrm{d}}(O_i), \ldots, \widehat{H}_{s,v_K}^{\mathrm{d}}(O_i)\right)^\top.$$

The targeting step proceeds backward in time. Let

$$\widehat{Y}_{\tau+1,i}^{\mathrm{d},*} = Y_i.$$

For $s = \tau, \tau-1, \ldots, 0$, fit a fluctuation submodel for the current outcome regression using $\widehat{Y}_{s+1,i}^{\mathrm{d},*}$ as the outcome, $\widehat{m}_{s,i}^{\mathrm{d}}(A_{s,i}, H_{s,i})$ as the initial prediction, and $\widehat{H}_s^{\mathrm{d}}(O_i)$ as the clever covariate. With an identity link, the fluctuation model is

$$\widehat{m}_{s,\epsilon}^{\mathrm{d}}(A_{s,i}, H_{s,i}) = \widehat{m}_s^{\mathrm{d}}(A_{s,i}, H_{s,i}) + \epsilon_s^\top \widehat{H}_s^{\mathrm{d}}(O_i),$$

where $\epsilon_s \in \mathbb{R}^K$. With a bounded outcome and logit link, the fluctuation model is

$$\operatorname{logit}\widehat{m}_{s,\epsilon}^{\mathrm{d}}(A_{s,i}, H_{s,i}) = \operatorname{logit}\widehat{m}_s^{\mathrm{d}}(A_{s,i}, H_{s,i}) + \epsilon_s^\top \widehat{H}_s^{\mathrm{d}}(O_i).$$

Let $\widehat{\epsilon}_s$ denote the fitted fluctuation coefficient and write

$$\widehat{m}_s^{\mathrm{d},*} = \widehat{m}_{s,\widehat{\epsilon}_s}^{\mathrm{d}}.$$

After updating the regression at time s, define the next outcome for the backward recursion by evaluating the updated regression under the modified treatment value:

$$\widehat{Y}_{s,i}^{\mathrm{d},*} = \widehat{m}_s^{\mathrm{d},*}(A_{s,i}^{\mathrm{d}}, H_{s,i}).$$

This updated pseudo-outcome is then used in the fluctuation at time $s-1$.

The subgroup TMLE of the policy-specific conditional mean is the plug-in estimator

$$\widehat{\theta}_{\mathrm{d},\mathrm{tmle}}(v) = \mathbb{P}_n\left[\frac{\mathbb{1}(V=v)}{\widehat{p}_v}\widehat{m}_0^{\mathrm{d},*}(A_0^{\mathrm{d}}, L_0)\right].$$

For two policies d_1 and d_0, the contrast estimator is

$$\widehat{\tau}_{\mathrm{tmle}}(v) = \widehat{\theta}_{\mathrm{d}_1,\mathrm{tmle}}(v) - \widehat{\theta}_{\mathrm{d}_0,\mathrm{tmle}}(v).$$

The estimated efficient influence function for the policy-specific subgroup mean is

$$\widehat{D}_{\mathrm{d},v}(O_i) = \frac{\mathbb{1}(V_i = v)}{\widehat{p}_v}\left[\widehat{m}_0^{\mathrm{d},*}(A_{0,i}^{\mathrm{d}}, L_{0,i}) - \widehat{\theta}_{\mathrm{d},\mathrm{tmle}}(v) + \sum_{s=0}^{\tau}\widehat{\omega}_{s,i}^{\mathrm{d}}\widehat{\Delta}_{s,i}^{\mathrm{d},*}\right],$$

where

$$\widehat{\Delta}_{s,i}^{\mathrm{d},*} = \widehat{m}_{s+1}^{\mathrm{d},*}(A_{s+1,i}^{\mathrm{d}}, H_{s+1,i}) - \widehat{m}_s^{\mathrm{d},*}(A_{s,i}, H_{s,i}), \qquad s = 0, \ldots, \tau - 1,$$

and

$$\widehat{\Delta}_{\tau,i}^{\mathrm{d},*} = Y_i - \widehat{m}_{\tau}^{\mathrm{d},*}(A_{\tau,i}, H_{\tau,i}).$$

For a contrast, define

$$\widehat{D}_{\tau,v}(O_i) = \widehat{D}_{\mathrm{d}_1,v}(O_i) - \widehat{D}_{\mathrm{d}_0,v}(O_i).$$

For a single subgroup v, the estimated standard error of $\widehat{\tau}_{\mathrm{tmle}}(v)$ is

$$\widehat{\mathrm{se}}\{\widehat{\tau}_{\mathrm{tmle}}(v)\} = \left[\frac{1}{n}\mathbb{P}_n\left\{\widehat{D}_{\tau,v}(O)^2\right\}\right]^{1/2}.$$

For simultaneous inference over $v \in \mathcal{V}$, define the empirical covariance matrix

$$\widehat{\Sigma}_{\tau} = \mathbb{P}_n\left[\widehat{D}_{\tau}(O)\widehat{D}_{\tau}(O)^{\top}\right],$$

where

$$\widehat{D}_{\tau}(O) = \left(\widehat{D}_{\tau,v_1}(O), \ldots, \widehat{D}_{\tau,v_K}(O)\right)^{\top}.$$

Then

$$\sqrt{n}\left\{\widehat{\tau}_{\mathrm{tmle}} - \tau\right\}$$

is approximated by a mean-zero multivariate normal distribution with covariance matrix $\widehat{\Sigma}_{\tau}$. Simultaneous confidence intervals may be formed using the quantiles of

$$\max_{v\in\mathcal{V}}\left|Z_v/\widehat{\sigma}_v\right|, \qquad Z \sim N(0, \widehat{\Sigma}_{\tau}),$$

where $\widehat{\sigma}_v^2$ is the vth diagonal element of $\widehat{\Sigma}_{\tau}$.

Remark 5. The targeting step above updates the full vector of subgroup parameters jointly at each time point. This is preferable to fitting a separate TMLE within each subgroup when some subgroups are small, because the nuisance regressions are estimated using the full sample while the clever covariates target the subgroup-specific components of the efficient influence function.

3.3. DR-Learner for conditional LMTP effects

When V is continuous or high-dimensional, the target

$$v \mapsto \theta_{\mathrm{d}}(v)$$

is a conditional mean function rather than a finite-dimensional parameter. The pseudo-outcome representation in Theorem 2 suggests a direct meta-learning approach: estimate the uncentered influence-function pseudo-outcome and regress it on V.

For a single policy d, the DR-Learner proceeds as follows.

1. Split the sample into folds $\mathcal{I}_1, \ldots, \mathcal{I}_J$.
2. On each training sample $\mathcal{I}_{-j}$, estimate the nuisance functions
$$\eta^{\mathrm{d}} = (m_0^{\mathrm{d}}, \ldots, m_\tau^{\mathrm{d}}, r_0^{\mathrm{d}}, \ldots, r_\tau^{\mathrm{d}}).$$
3. For each observation $i \in \mathcal{I}_j$, construct the cross-fitted pseudo-outcome
$$\widehat{\varphi}_{\mathrm{d},i} = \widehat{m}_{0,i}^{\mathrm{d}}(A_{0,i}^{\mathrm{d}}, L_{0,i}) + \sum_{s=0}^{\tau} \widehat{\omega}_{s,i}^{\mathrm{d}} \widehat{\Delta}_{s,i}^{\mathrm{d}}.$$
4. Regress $\widehat{\varphi}_{\mathrm{d},i}$ on V_i using a chosen second-stage regression method. The fitted regression function is
$$\widehat{\theta}_{\mathrm{d},\mathrm{dr}}(v) = \widehat{\mathbb{E}}_n \left[\widehat{\varphi}_{\mathrm{d}} \mid V = v \right].$$

For two policies d_1 and d_0, construct the contrast pseudo-outcome
$$\widehat{\varphi}_{\tau,i} = \widehat{\varphi}_{\mathrm{d}_1,i} - \widehat{\varphi}_{\mathrm{d}_0,i},$$
and regress it on V_i:
$$\widehat{\tau}_{\mathrm{dr}}(v) = \widehat{\mathbb{E}}_n \left[\widehat{\varphi}_{\tau} \mid V = v \right].$$
Equivalently,
$$\widehat{\tau}_{\mathrm{dr}}(v) = \widehat{\theta}_{\mathrm{d}_1,\mathrm{dr}}(v) - \widehat{\theta}_{\mathrm{d}_0,\mathrm{dr}}(v),$$
if the same second-stage regression method is applied separately to the two policy-specific pseudo-outcomes.

The second-stage regression may be chosen to reflect the scientific structure expected in the conditional effect. If V is low-dimensional and continuous, local linear regression, splines, generalized additive models, or local polynomial regression are natural choices. If V is multivariate, random forests, gradient boosting, neural networks, or Super Learner ensembles may be used. If the scientific question concerns a pre-specified low-dimensional projection of the conditional effect, one may regress the pseudo-outcome on a parametric or semiparametric working model for V.

The corresponding benchmark estimator is the regression that would be obtained if the nuisance functions were known:
$$\widetilde{\theta}_{\mathrm{d}}(v) = \widehat{\mathbb{E}}_n \left[\varphi_{\mathrm{d}}(O; \eta^{\mathrm{d}}) \mid V = v \right],$$
and, for a contrast,
$$\widetilde{\tau}(v) = \widehat{\mathbb{E}}_n \left[\varphi_{\tau}(O; \eta) \mid V = v \right].$$
The theoretical results in Section 4 compare the feasible DR-Learner to this benchmark regression. In particular, oracle equivalence holds when the smoothed second-order bias induced by nuisance estimation is negligible relative to the oracle second-stage regression error.

Remark 6. For finite discrete V, the DR-Learner with a saturated second-stage regression is simply the subgroup mean of the estimated pseudo-outcomes. Thus the finite-subgroup estimator can be viewed as the discrete special case of the pseudo-outcome regression strategy. The targeted estimator, however, has the additional advantage that it updates the nuisance regressions to solve the efficient influence-function equations for the subgroup parameters.

Remark 7 *(Second-stage regression and HAL).* Uncertainty quantification for the DR-Learner depends on both the second-stage regression method and the target of inference. For finite-dimensional projections of the conditional effect, standard asymptotic linearity can often be obtained from the corresponding projected estimating equations. For pointwise or uniform inference on a nonparametric conditional effect curve, additional analysis is required.

One strategically attractive choice for the second-stage regression is the highly adaptive lasso (HAL). HAL estimates cadlag functions with finite sectional variation norm by empirical risk minimization over a large dictionary of tensor-product indicator basis functions subject to an ℓ_1 constraint. For regression-type function estimation, HAL attains loss-based convergence rates that typically imply $L_2(P_0)$ convergence at rate $n^{-1/3}$, up to logarithmic factors, under only a finite sectional variation norm condition. This rate is faster than $n^{-1/4}$, which is often the critical threshold appearing in second-order remainder conditions for doubly robust and targeted estimators. HAL is therefore appealing as a second-stage learner because it can estimate flexible conditional effect functions while retaining nonparametric rate guarantees under relatively weak global regularity assumptions.

Undersmoothed HAL may be particularly relevant if the goal is inference rather than only prediction. Existing work shows that undersmoothed HAL can yield asymptotically efficient plug-in estimators of pathwise differentiable features of an infinite-dimensional regression function, and higher-order spline extensions of HAL provide pointwise asymptotic normality and uniform convergence rates for function estimation under stronger smoothness conditions. These results suggest that HAL-based second-stage regression may provide a promising route toward inference for conditional LMTP effect curves. However, establishing pointwise or uniform confidence bands for the LMTP DR-Learner requires theory specific to the pseudo-outcome, the second-stage target, and the chosen smoothing or undersmoothing rule. We therefore establish a general oracle-equivalence result for the DR-Learner in Section 4 and leave method-specific pointwise and uniform inference for conditional LMTP effect curves to future work.

3.4. Projections onto working marginal structural models

The subgroup TMLE of Section 3.2 and the DR-Learner of Section 3.3 occupy two extremes: fully saturated inference over a finite $\mathcal{V}$, and fully nonparametric estimation of $v \mapsto \theta_{\mathrm{d}}(v)$ at nonparametric rates. Applied reporting often calls for something in between — a low-dimensional, interpretable summary of how the policy effect varies with V, with root-n inference — or for pointwise and uniform inference on the conditional effect curve itself. Both are obtained by regressing the influence-function pseudo-outcome on a basis in V: a *fixed* basis yields a working marginal structural model (MSM) whose projection coefficients are pathwise differentiable, while a basis of *growing* dimension, undersmoothed so that approximation bias is negligible, yields valid inference on θ_{d} itself. We present the two regimes together because they share one estimator.

3.5. *Fixed-dimensional working models*

Let $b(v) = \{b_1(v), \ldots, b_k(v)\}^\top$ be a fixed, user-specified vector of basis functions — for example $b(v) = (1, v)^\top$ for a linear trend in a scalar modifier, or a coarse spline basis — and let $h(v) \geq 0$ be a user-specified weight function, with $h \equiv 1$ as the default. Define the working-model projection of the conditional LMTP mean,

$$\beta_{\mathrm{d}} = \arg\min_{\beta \in \mathbb{R}^k} \mathbb{E}\left[h(V)\left\{\theta_{\mathrm{d}}(V) - b(V)^\top \beta\right\}^2\right] = M^{-1}\,\mathbb{E}\left\{h(V)\,b(V)\,\theta_{\mathrm{d}}(V)\right\}, \qquad M = \mathbb{E}\left\{h(V)\,b(V)\,b(V)^\top\right\},$$

assuming M is nonsingular. The parameter β_{d} is defined without assuming the working model is correct: $v \mapsto b(v)^\top \beta_{\mathrm{d}}$ is the weighted least-squares projection of θ_{d} onto the span of b, and retains this interpretation under arbitrary misspecification **??**. When the working model happens to hold, $b(v)^\top \beta_{\mathrm{d}} = \theta_{\mathrm{d}}(v)$. For a contrast of two policies, $\beta_\tau = \beta_{\mathrm{d}_1} - \beta_{\mathrm{d}_0}$ is the projection of $\tau(\cdot)$, obtained by replacing θ_{d} with τ throughout, or equivalently by using the contrast pseudo-outcome $\varphi_\tau = \varphi_{\mathrm{d}_1} - \varphi_{\mathrm{d}_0}$.

Because $\mathbb{E}\{\varphi_{\mathrm{d}}(O; \eta_0^{\mathrm{d}}) \mid V = v\} = \theta_{\mathrm{d}}(v)$ by Theorem 2, the projection coefficients admit the moment representation $\beta_{\mathrm{d}} = M^{-1}\mathbb{E}\{h(V)b(V)\varphi_{\mathrm{d}}(O; \eta_0^{\mathrm{d}})\}$, which is a smooth finite-dimensional functional of the observed data law.

PROPOSITION 1 (EFFICIENT INFLUENCE FUNCTION OF THE PROJECTION). *Under Assumptions A1–A5 and nonsingular M, the parameter β_{d} is pathwise differentiable with efficient influence function*

$$D_{\beta,\mathrm{d}}(O) = M^{-1}\,h(V)\,b(V)\left\{\varphi_{\mathrm{d}}(O; \eta_0^{\mathrm{d}}) - b(V)^\top \beta_{\mathrm{d}}\right\}.$$

For the contrast, $D_{\beta,\tau} = D_{\beta,\mathrm{d}_1} - D_{\beta,\mathrm{d}_0}$.

Proof. The parameter solves the moment equation $\mathbb{E}\,U(O;\beta) = 0$ with $U(O;\beta) = h(V)b(V)\{\varphi_{\mathrm{d}}(O; \eta_0^{\mathrm{d}}) - b(V)^\top\beta\}$. The functional $P \mapsto \mathbb{E}_P\{h(V)b(V)\theta_{\mathrm{d},P}(V)\}$ has efficient influence function $h(V)b(V)\varphi_{\mathrm{d}}(O; \eta_0^{\mathrm{d}})$ minus its mean, since φ_{d} is the uncentered efficient influence function for the LMTP mean and $h\,b$ is a fixed baseline-measurable weight; the contribution of the marginal law of V through M is absorbed by centering at $b(V)^\top\beta_{\mathrm{d}}$, which renders $D_{\beta,\mathrm{d}}$ mean zero by the normal equations. The delta method for estimating equations, with derivative matrix $\partial_\beta\,\mathbb{E}\,U = -M$, gives the stated form. Details are provided in the Supplementary Material. □

The estimator is the weighted least-squares regression of the cross-fitted pseudo-outcomes of Section 3.1 on the basis:

$$\widehat{\beta}_{\mathrm{d}} = \widehat{M}^{-1}\,\mathbb{P}_n\left\{h(V)\,b(V)\,\widehat{\varphi}_{\mathrm{d}}(O)\right\}, \qquad \widehat{M} = \mathbb{P}_n\left\{h(V)\,b(V)\,b(V)^\top\right\}.$$

THEOREM 3 (ASYMPTOTIC LINEARITY OF THE WORKING-MODEL PROJECTION). *Suppose Assumptions A1–A5 hold, M is nonsingular, h and b are bounded, the pseudo-outcomes are constructed with cross-fitting, the cumulative estimated ratios are bounded in probability, and the nuisance product-rate condition*

$$\sum_{s=0}^{\tau} \left\|\widehat{r}_s^{\mathrm{d}} - r_s^{\mathrm{d}}\right\|_2 \left\|\widehat{m}_s^{\mathrm{d}} - m_s^{\mathrm{d}}\right\|_2 = o_p(n^{-1/2})$$

holds in the marginal $L_2(P_0)$ norm. Then

$$\widehat{\beta}_{\mathrm{d}} - \beta_{\mathrm{d}} = \mathbb{P}_n D_{\beta,\mathrm{d}} + o_p(n^{-1/2}), \qquad \sqrt{n}\left(\widehat{\beta}_{\mathrm{d}} - \beta_{\mathrm{d}}\right) \rightsquigarrow N(0, \Sigma_\beta), \qquad \Sigma_\beta = \mathbb{E}\left\{D_{\beta,\mathrm{d}} D_{\beta,\mathrm{d}}^\top\right\},$$

and Σ_β is consistently estimated by the sandwich estimator

$$\widehat{\Sigma}_\beta = \widehat{M}^{-1}\,\mathbb{P}_n\left[h(V)^2\,b(V)\,b(V)^\top\left\{\widehat{\varphi}_{\mathrm{d}}(O) - b(V)^\top\widehat{\beta}_{\mathrm{d}}\right\}^2\right]\widehat{M}^{-1}.$$

The same statements hold for the contrast with φ_τ, $D_{\beta,\tau}$, and β_τ.

Proof. Writing $\widehat{\beta}_{\mathrm{d}} - \beta_{\mathrm{d}} = \widehat{M}^{-1}\mathbb{P}_n\,h\,b\,\{\widehat{\varphi}_{\mathrm{d}} - b^\top\beta_{\mathrm{d}}\}$ and decomposing $\widehat{\varphi}_{\mathrm{d}} = \varphi_{\mathrm{d}}(O;\eta_0^{\mathrm{d}}) + b^{\mathrm{d}}_{\widehat{\eta}}(V) + e_{\widehat{\eta}}(O)$ as in the proof of Theorem 5, the term involving $\varphi_{\mathrm{d}}(O;\eta_0^{\mathrm{d}})$ yields the influence-function expansion, the conditionally mean-zero term $e_{\widehat{\eta}}$ is $o_p(n^{-1/2})$ by cross-fitting, and the bias term is bounded by $\|h\,b\|_\infty\,\mathbb{E}|b^{\mathrm{d}}_{\widehat{\eta}}(V)|$, which is $o_p(n^{-1/2})$ under the product-rate condition by Proposition 3 and the Cauchy–Schwarz inequality; $\widehat{M} \to M$ in probability completes the argument. Details are given in the Supplementary Material. □

Note that the rate condition in Theorem 3 involves the *marginal* $L_2(P_0)$ norms rather than the subgroup-restricted norms of Corollary 2: because the projection averages over V, no subgroup-specific nuisance accuracy is required.

Remark 8 *(Simultaneous bands for the fitted summary).* Wald intervals for individual coefficients (for example, a linear-trend slope) follow directly. For the fitted curve, a simultaneous band over any finite grid $v_1, \ldots, v_G$ is $b(v_g)^\top\widehat{\beta}_{\mathrm{d}} \pm c_{1-\alpha}\{b(v_g)^\top\widehat{\Sigma}_\beta\,b(v_g)/n\}^{1/2}$, where $c_{1-\alpha}$ is the max-$|Z|$ critical value computed from the Gaussian vector with correlation matrix induced by $\{b(v_g)^\top\widehat{\Sigma}_\beta\,b(v_{g'})\}_{g,g'}$, exactly as in Section 3.2. Homogeneity of the effect in V within the working model corresponds to the null that all non-intercept coordinates of β_τ vanish, testable by a Wald statistic based on $\widehat{\Sigma}_\beta$.

Remark 9 *(The subgroup TMLE as a saturated projection).* When V is discrete with support $\{v_1, \ldots, v_K\}$, taking $h \equiv 1$ and the saturated basis $b(v) = \{\mathbb{1}(v = v_1), \ldots, \mathbb{1}(v = v_K)\}^\top$ gives $\widehat{\beta}_{\mathrm{d},k} = \mathbb{P}_n\{\mathbb{1}(V = v_k)\widehat{\varphi}_{\mathrm{d}}\}/\widehat{p}_{v_k}$, the one-step subgroup estimator, with $D_{\beta,\mathrm{d}}$ reducing coordinatewise to the subgroup efficient influence functions $D_{\mathrm{d},v}$ of Section 3.2. The working-model formulation therefore nests the subgroup analysis, and the TMLE of Section 3.2 may be viewed as its targeted refinement, whose plug-in form guarantees that estimates respect the parameter space.

3.6. Undersmoothed B-spline sieves

When V is continuous and the target is $\theta_{\mathrm{d}}(v)$ itself, the same weighted least-squares estimator may be used with a basis whose dimension grows with n. Let V have compact support $\mathcal{V} \subset \mathbb{R}^{d_V}$, and let $b_{k_n}(v) \in \mathbb{R}^{k_n}$ collect tensor-product B-splines of fixed order ℓ on a quasi-uniform knot sequence, with $k_n \to \infty$. Write $\widehat{\theta}_{\mathrm{d},n}(v) = b_{k_n}(v)^\top\widehat{\beta}_{\mathrm{d},n}$ for the resulting series estimator and $\sigma_n^2(v) = b_{k_n}(v)^\top\Sigma_{\beta,n}\,b_{k_n}(v)/n$ for its pointwise variance, with $\Sigma_{\beta,n}$ as in Theorem 3 at dimension k_n. For local bases such as B-splines, $\sigma_n(v) \asymp (k_n/n)^{1/2}$, while the sup-norm approximation bias of the projection satisfies $\sup_v |b_{k_n}(v)^\top\beta_{\mathrm{d},n} - \theta_{\mathrm{d}}(v)| = O(k_n^{-s/d_V})$ when θ_{d} is s-smooth with $s \le \ell$. *Undersmoothing* selects k_n large enough that the bias is negligible relative to the pointwise standard deviation, so that the confidence statements center on θ_{d} rather than on its projection.

Assumption A6 *(Sieve regularity).* (i) The density of V is bounded away from zero and infinity on $\mathcal{V}$; (ii) θ_{d} belongs to a Hölder class of smoothness $s \le \ell$ on $\mathcal{V}$; (iii) the dimension satisfies $k_n \log n/n \to 0$ and the undersmoothing condition $n\,k_n^{-(2s/d_V+1)} \to 0$; (iv) the pseudo-outcome has uniformly bounded conditional $(2+\varepsilon)$-th moments given

V, as holds under condition 3 of Assumption A7; and (v) the nuisance product-rate condition of Theorem 3 holds.

Condition (iii) is nonvacuous whenever $n^{d_V/(2s+d_V)} \ll k_n \ll n/\log n$: the sieve dimension is taken of larger order than the mean-squared-error-optimal choice, sacrificing rate optimality for centering. Under (i), B-spline Gram matrices have eigenvalues bounded away from zero and infinity and $\sup_v \|b_{k_n}(v)\| \asymp k_n^{1/2}$, so the conditions of the least-squares series literature apply **??**.

PROPOSITION 2 (POINTWISE AND UNIFORM SIEVE INFERENCE). *Under Assumptions A1–A5 and A6, with cross-fitted pseudo-outcomes,*

$$\frac{\widehat{\theta}_{\mathrm{d},n}(v) - \theta_{\mathrm{d}}(v)}{\widehat{\sigma}_n(v)} \rightsquigarrow N(0,1) \quad \textit{for each fixed } v,$$

and the multiplier-bootstrap critical value $\widehat{c}_{1-\alpha}$ *for* $\sup_{v\in\mathcal{V}} |\widehat{\theta}_{\mathrm{d},n}(v) - \theta_{\mathrm{d}}(v)|/\widehat{\sigma}_n(v)$ *yields uniformly valid confidence bands,*

$$\mathbb{P}\left\{\theta_{\mathrm{d}}(v) \in \widehat{\theta}_{\mathrm{d},n}(v) \pm \widehat{c}_{1-\alpha}\,\widehat{\sigma}_n(v) \textit{ for all } v \in \mathcal{V}\right\} \to 1-\alpha.$$

The analogous statements hold for the contrast $\tau(\cdot)$.

Proof. The result follows by verifying the conditions of **?** and **?** for least-squares series regression with the estimated outcome $\widehat{\varphi}_{\mathrm{d}}$: the decomposition $\widehat{\varphi}_{\mathrm{d}} = \varphi_{\mathrm{d}}(O;\eta_0^{\mathrm{d}}) + b_{\widehat{\eta}}^{\mathrm{d}}(V) + e_{\widehat{\eta}}(O)$ reduces the problem to the oracle series regression plus a nuisance-bias contribution whose projection is uniformly $o_p\{(k_n/n)^{1/2}\}$ under condition (v), since the product-rate condition implies $\|b_{\widehat{\eta}}^{\mathrm{d}}\|_2 = o_p(n^{-1/2}) = o_p\{(k_n/n)^{1/2}\}$, and a conditionally centered contribution controlled by cross-fitting. The undersmoothing condition (iii) renders the approximation bias $o_p\{\sigma_n(v)\}$ uniformly. Verification of the technical conditions is given in the Supplementary Material. □

Remark 10 *(Nuisance rates are again weaker than root-n).* As in Remark 14, requiring the nuisance products to be $o_p(n^{-1/2})$ in condition (v) is convenient but conservative: since the sieve standard deviation is of order $(k_n/n)^{1/2} \gg n^{-1/2}$, it suffices that the products be $o_p\{(k_n/n)^{1/2}\}$, a strictly weaker requirement that grows weaker as the band is made more flexible.

Remark 11 *(Practical selection of the sieve).* In practice we take cubic B-splines ($\ell = 4$), place knots at quantiles of V, select a mean-squared-error-optimal dimension $\widehat{k}$ by cross-validation on the pseudo-outcome regression, and then undersmooth by a fixed multiple, e.g. $k_n = \lceil c\,\widehat{k}\rceil$ with $c \in [1.5, 2]$; the resulting bands are wider than the cross-validated fit's naive bands, which is the price of centering on θ_{d} rather than a smoothed surrogate. Alternatives that avoid undersmoothing — explicit bias correction in the second-stage regression **?** — apply here as well by substituting our pseudo-outcome, exactly as in the point-treatment setting **?**.

4. THEORETICAL PROPERTIES

We now give large-sample properties of the proposed estimators. The finite subgroup TMLE is a regular asymptotically linear estimator of the vector of subgroup-specific LMTP effects under standard cross-fitting and second-order remainder conditions. The DR-Learner is analyzed as a regression with estimated pseudo-outcomes. Its behavior is

compared to the benchmark regression that would be obtained if the nuisance functions were known.

Throughout this section, let P_0 denote the true observed data law. For a policy $\mathbb{d}$, let $\eta_0^{\mathbb{d}}$ denote the corresponding true nuisance functions, and let

$$\varphi_{\mathbb{d}}(O; \eta_0^{\mathbb{d}})$$

denote the uncentered LMTP influence-function pseudo-outcome defined in Section 2.6. For a contrast of two policies $\mathbb{d}_1$ and $\mathbb{d}_0$, define

$$\varphi_\tau(O; \eta_0) = \varphi_{\mathbb{d}_1}(O; \eta_0^{\mathbb{d}_1}) - \varphi_{\mathbb{d}_0}(O; \eta_0^{\mathbb{d}_0}).$$

4.1. Asymptotic linearity of the subgroup TMLE

Suppose V is discrete with finite support

$$\mathcal{V} = \{v_1, \ldots, v_K\}.$$

Let

$$\theta_{\mathbb{d}} = \{\theta_{\mathbb{d}}(v_1), \ldots, \theta_{\mathbb{d}}(v_K)\}^\top$$

denote the vector of policy-specific subgroup LMTP means. For a contrast of two policies, write

$$\tau = \{\tau(v_1), \ldots, \tau(v_K)\}^\top.$$

The efficient influence function for the policy-specific subgroup mean $\theta_{\mathbb{d}}(v)$ is

$$D_{\mathbb{d},v}(O) = \frac{\mathbb{1}(V = v)}{p_v} \left\{ \varphi_{\mathbb{d}}(O; \eta_0^{\mathbb{d}}) - \theta_{\mathbb{d}}(v) \right\}, \qquad p_v = P_0(V = v).$$

For a contrast,

$$D_{\tau,v}(O) = D_{\mathbb{d}_1,v}(O) - D_{\mathbb{d}_0,v}(O).$$

Let

$$D_\tau(O) = \{D_{\tau,v_1}(O), \ldots, D_{\tau,v_K}(O)\}^\top.$$

Assumption A7 *(Regularity for subgroup TMLE).* The following conditions hold.

1. The identification assumptions in Section 2.4 hold for each policy under consideration.
2. For each $v \in \mathcal{V}$, $p_v > 0$, and $\min_{v \in \mathcal{V}} p_v$ is bounded away from zero.
3. The outcome and sequential regressions are uniformly bounded, and the cumulative density ratios $\omega_s^{\mathbb{d}}(O) = \prod_{t=0}^{s} r_t^{\mathbb{d}}(A_t, H_t)$ are uniformly bounded or have sufficiently many bounded moments.
4. The nuisance functions are estimated using cross-fitting, so that nuisance estimation and evaluation of the estimating equations are performed on independent samples.
5. The targeting step solves the empirical efficient influence-function equations up to negligible error:
$$\mathbb{P}_n \widehat{D}_{\mathbb{d},v} = o_p(n^{-1/2}) \qquad \text{for all } v \in \mathcal{V}.$$
6. The second-order remainder for each policy satisfies
$$R_{\mathbb{d},n}(v) = o_p(n^{-1/2}) \qquad \text{for all } v \in \mathcal{V}.$$

THEOREM 4 (ASYMPTOTIC LINEARITY OF THE SUBGROUP TMLE). *Under Assumption A7,*

$$\widehat{\theta}_{\mathbb{d},\mathrm{tmle}}(v) - \theta_{\mathbb{d}}(v) = \mathbb{P}_n D_{\mathbb{d},v} + o_p(n^{-1/2})$$

for each $v \in \mathcal{V}$. *Hence*

$$\sqrt{n}\left\{\widehat{\theta}_{\mathbb{d},\mathrm{tmle}} - \theta_{\mathbb{d}}\right\} \rightsquigarrow N(0, \Sigma_{\mathbb{d}}),$$

where

$$\Sigma_{\mathbb{d}} = \mathbb{E}_{P_0}\left[D_{\mathbb{d}}(O) D_{\mathbb{d}}(O)^\top\right].$$

For a contrast of two policies,

$$\widehat{\tau}_{\mathrm{tmle}}(v) - \tau(v) = \mathbb{P}_n D_{\tau,v} + o_p(n^{-1/2}),$$

and therefore

$$\sqrt{n}\left\{\widehat{\tau}_{\mathrm{tmle}} - \tau\right\} \rightsquigarrow N(0, \Sigma_\tau),$$

where

$$\Sigma_\tau = \mathbb{E}_{P_0}\left[D_\tau(O) D_\tau(O)^\top\right].$$

Proof. We give the argument for a single policy. The contrast result follows by linearity. By the definition of the subgroup parameter and the efficient influence function,

$$\widehat{\theta}_{\mathbb{d},\mathrm{tmle}}(v) - \theta_{\mathbb{d}}(v) = (\mathbb{P}_n - P_0) D_{\mathbb{d},v} + \mathbb{P}_n \widehat{D}_{\mathbb{d},v} + R_{\mathbb{d},n}(v) + o_p(n^{-1/2}).$$

The empirical estimating equation term $\mathbb{P}_n \widehat{D}_{\mathbb{d},v}$ is $o_p(n^{-1/2})$ by the targeting condition. The second-order remainder is $o_p(n^{-1/2})$ by assumption. The cross-fitting construction removes the need for Donsker restrictions on the estimated nuisance functions and allows the leading empirical process term to be evaluated at the fixed true influence function. Thus

$$\widehat{\theta}_{\mathbb{d},\mathrm{tmle}}(v) - \theta_{\mathbb{d}}(v) = \mathbb{P}_n D_{\mathbb{d},v} + o_p(n^{-1/2}).$$

The multivariate central limit theorem gives the joint normal limit over $v \in \mathcal{V}$. □

Remark 12. The empirical covariance estimator

$$\widehat{\Sigma}_\tau = \mathbb{P}_n\left[\widehat{D}_\tau(O) \widehat{D}_\tau(O)^\top\right]$$

is consistent for Σ_τ under the same conditions, provided the estimated influence functions converge in $L_2(P_0)$ to their population limits.

4.2. Oracle equivalence of the DR-Learner

We next study the DR-Learner for the conditional LMTP contrast

$$\tau(v) = \mathbb{E}_{P_0}\{\varphi_\tau(O; \eta_0) \mid V = v\}.$$

Let $\widehat{\varphi}_{\tau,i}$ be the cross-fitted estimated pseudo-outcome defined in Section 3.3. Let

$$\widehat{\mathbb{E}}_n[Z \mid V = v]$$

denote a generic second-stage regression estimator applied to an outcome Z and evaluated at $V = v$. This notation may represent, for example, a series regression, spline regression,

local polynomial regression, random forest, or any other regression method used in the second stage.

The feasible DR-Learner is

$$\widehat{\tau}_{\mathrm{dr}}(v) = \widehat{\mathbb{E}}_n \left[\widehat{\varphi}_\tau \mid V = v \right].$$

The benchmark regression that would be obtained if the nuisance functions were known is

$$\widetilde{\tau}(v) = \widehat{\mathbb{E}}_n \left[\varphi_\tau(O; \eta_0) \mid V = v \right].$$

This benchmark is not available in practice; it is used only to measure the cost of estimating the nuisance functions.

Define the conditional pseudo-outcome bias

$$b_{\widehat{\eta}}(v) = \mathbb{E}_{P_0} \left[\widehat{\varphi}_\tau(O) - \varphi_\tau(O; \eta_0) \mid \widehat{\eta}, V = v \right].$$

Equivalently, $b_{\widehat{\eta}}(v)$ is the regression function of the pseudo-outcome estimation error on V, conditional on the nuisance training sample.

Assumption A8 *(Second-stage stability).* We say that $\widehat{\mathbb{E}}_n$ is stable at $V = v$, with respect to a distance d, if

$$\frac{\widehat{\mathbb{E}}_n \left[\widehat{\varphi}_\tau(O) \mid V = v \right] - \widehat{\mathbb{E}}_n \left[\varphi_\tau(O; \eta_0) \mid V = v \right] - \widehat{\mathbb{E}}_n \left[b_{\widehat{\eta}}(V) \mid V = v \right]}{\sqrt{\mathbb{E}\left[\left(\widehat{\mathbb{E}}_n \left[\varphi_\tau(O; \eta_0) \mid V = v \right] - \mathbb{E}_{P_0} \left[\varphi_\tau(O; \eta_0) \mid V = v \right] \right)^2 \right]}} \xrightarrow{p} 0$$

whenever

$$d\{\widehat{\varphi}_\tau, \varphi_\tau(\eta_0)\} \xrightarrow{p} 0.$$

Theorem 5 (Oracle equivalence of the LMTP DR-Learner). *Suppose the identification assumptions hold for both policies* d_1 *and* d_0*. Suppose the nuisance functions are estimated with cross-fitting and that Assumption A8 holds. Then*

$$\widehat{\tau}_{\mathrm{dr}}(v) - \widetilde{\tau}(v) = \widehat{\mathbb{E}}_n \left[b_{\widehat{\eta}}(V) \mid V = v \right] + o_p\{R_n^\star(v)\}.$$

Consequently,

$$\widehat{\tau}_{\mathrm{dr}}(v) - \tau(v) = \widetilde{\tau}(v) - \tau(v) + \widehat{\mathbb{E}}_n \left[b_{\widehat{\eta}}(V) \mid V = v \right] + o_p\{R_n^\star(v)\}.$$

If

$$\widehat{\mathbb{E}}_n \left[b_{\widehat{\eta}}(V) \mid V = v \right] = o_p\{R_n^\star(v)\},$$

then

$$\widehat{\tau}_{\mathrm{dr}}(v) - \widetilde{\tau}(v) = o_p\{R_n^\star(v)\}.$$

Thus the feasible DR-Learner is oracle equivalent at v*.*

Proof. Conditional on the nuisance training sample, decompose the estimated pseudo-outcome as

$$\widehat{\varphi}_\tau(O) = \varphi_\tau(O; \eta_0) + b_{\widehat{\eta}}(V) + e_{\widehat{\eta}}(O),$$

where

$$\mathbb{E}_{P_0}\{e_{\widehat{\eta}}(O) \mid \widehat{\eta}, V\} = 0.$$

Applying the second-stage regression estimator gives

$$\widehat{\tau}_{\mathrm{dr}}(v) = \widehat{\mathbb{E}}_n\left[\varphi_\tau(O;\eta_0) + b_{\widehat{\eta}}(V) + e_{\widehat{\eta}}(O) \mid V = v\right].$$

By Assumption A8,

$$\widehat{\tau}_{\mathrm{dr}}(v) - \widehat{\mathbb{E}}_n\left[\varphi_\tau(O;\eta_0) \mid V = v\right] = \widehat{\mathbb{E}}_n\left[b_{\widehat{\eta}}(V) \mid V = v\right] + o_p\{R_n^\star(v)\}.$$

Since

$$\widetilde{\tau}(v) = \widehat{\mathbb{E}}_n\left[\varphi_\tau(O;\eta_0) \mid V = v\right],$$

the first equation of the theorem follows. Adding and subtracting $\tau(v)$ gives the second second equation. The oracle equivalence claim follows if the smoothed conditional bias is negligible relative to the oracle regression error. □

COROLLARY 1 (CONSISTENCY OF THE DR-LEARNER). *Suppose the conditions of Theorem 5 hold. If the oracle regression is consistent for the truth, and the conditional pseudo-outcome bias decays to zero, then the oracle regression will be consistent. That is to say, If* $\widetilde{\tau}(v) - \tau(v) = o_p(1)$ *and* $\widehat{\mathbb{E}}_n\left[b_{\widehat{\eta}}(V) \mid V = v\right] = o_p(1)$, *then* $\widehat{\tau}_{\mathrm{dr}}(v) - \tau(v) = o_p(1)$.

Proof. The result follows immediately from the expansion in Theorem 5. □

4.3. Second-order bias and nuisance-rate conditions

Both estimators require control of a second-order remainder: condition 6 of Assumption A7 for the subgroup TMLE, and the conditional-bias condition $\widehat{\mathbb{E}}_n\{b_{\widehat{\eta}}(V) \mid V = v\} = o_p\{R_n^\star(v)\}$ of Theorem 5 for the DR-Learner. In this section we give these remainders an explicit form and derive sufficient nuisance rate conditions. The key object is the conditional bias of the uncentered pseudo-outcome under estimated nuisances,

$$b_{\widehat{\eta}}^{\mathbb{d}}(v) = \mathbb{E}_{P_0}\left\{\varphi_{\mathbb{d}}(O;\widehat{\eta}^{\mathbb{d}}) - \varphi_{\mathbb{d}}(O;\eta_0^{\mathbb{d}}) \mid V = v\right\},$$

where the expectation is over O with the estimated nuisances $\widehat{\eta}^{\mathbb{d}}$ held fixed (as guaranteed by cross-fitting). Because $\mathbb{E}_{P_0}\{\varphi_{\mathbb{d}}(O;\eta_0^{\mathbb{d}}) \mid V = v\} = \theta_{\mathbb{d}}(v)$ by Theorem 2, $b_{\widehat{\eta}}^{\mathbb{d}}$ is exactly the conditional analogue of the scalar second-order remainder, and the subgroup TMLE remainder satisfies $R_{\mathbb{d},n}(v) = b_{\widehat{\eta}}^{\mathbb{d}}(v) + o_p(n^{-1/2})$ under the conditions of Remark 15 below.

PROPOSITION 3 (PRODUCT FORM OF THE CONDITIONAL REMAINDER). *Let* $\widehat{\omega}_{s-1}^{\mathbb{d}}(O) = \prod_{k=0}^{s-1} \widehat{r}_k^{\mathbb{d}}(A_k, H_k)$ *with the convention* $\widehat{\omega}_{-1}^{\mathbb{d}} \equiv 1$. *Under Assumptions A1–A5, and provided the estimated ratios and regressions are uniformly bounded,*

$$b_{\widehat{\eta}}^{\mathbb{d}}(v) = -\sum_{s=0}^{\tau} \mathbb{E}_{P_0}\left[\widehat{\omega}_{s-1}^{\mathbb{d}}(O)\left\{\widehat{r}_s^{\mathbb{d}}(A_s,H_s) - r_s^{\mathbb{d}}(A_s,H_s)\right\}\left\{\widehat{m}_s^{\mathbb{d}}(A_s,H_s) - m_s^{\mathbb{d}}(A_s,H_s)\right\} \,\middle|\, V = v\right].$$

Proof. The argument telescopes the sum defining $\varphi_{\mathbb{d}}$ one time point at a time, using at each s the intervention-density identity $\mathbb{E}\{f(A_s^{\mathbb{d}}, H_s) \mid H_s\} = \mathbb{E}\{r_s^{\mathbb{d}}(A_s,H_s) f(A_s,H_s) \mid H_s\}$, which holds for any bounded f by Assumption A4 and the definition of $r_s^{\mathbb{d}}$. Note that the cumulative weight multiplying the s-th product is the *estimated* weight $\widehat{\omega}_{s-1}^{\mathbb{d}}$, not its population counterpart. Full details are given in the Supplementary Material. □

Two features of Proposition 3 drive the rate conditions used throughout Section 4. First, the remainder is a sum of $\tau+1$ terms, each a product of the estimation errors of one density ratio and its same-time-point sequential regression; there are no cross-time products. Second, each product enters only through its conditional mean given $V=v$, so all rates may be taken with respect to the subgroup-restricted norm

$$\|f\|_{2,v}^2 = \mathbb{E}_{P_0}\left\{f(O)^2 \mid V=v\right\}.$$

Corollary 2 (Sufficient product-rate conditions). *Suppose* $\max_s \sup_o |\widehat{\omega}_{s-1}^{\mathrm{d}}(o)| = O_p(1)$. *Then, by the Cauchy–Schwarz inequality,*

$$\left|b_{\widehat{\eta}}^{\mathrm{d}}(v)\right| \lesssim \sum_{s=0}^{\tau} \left\|\widehat{r}_s^{\mathrm{d}} - r_s^{\mathrm{d}}\right\|_{2,v} \left\|\widehat{m}_s^{\mathrm{d}} - m_s^{\mathrm{d}}\right\|_{2,v}.$$

In particular, if for every $s=0,\ldots,\tau$, $\|\widehat{r}_s^{\mathrm{d}} - r_s^{\mathrm{d}}\|_{2,v}\|\widehat{m}_s^{\mathrm{d}} - m_s^{\mathrm{d}}\|_{2,v} = o_p(n^{-1/2})$ — *as holds when each nuisance converges at the* $o_p(n^{-1/4})$ *rate attainable by many machine-learning estimators under smoothness or sparsity — then* $R_{\mathrm{d},n}(v) = o_p(n^{-1/2})$, *and condition 6 of Assumption A7 is satisfied.*

Remark 13 *(Multiple robustness).* The product form implies $2^{\tau+1}$-fold robustness: the conditional bias at v vanishes provided that, for *each* time point s separately, at least one of $\widehat{r}_s^{\mathrm{d}}$ and $\widehat{m}_s^{\mathrm{d}}$ is consistent in $\|\cdot\|_{2,v}$, with the other merely bounded. Consistent estimation of all ratios, or of all sequential regressions, are two special cases.

Remark 14 *(Weaker conditions for the DR-Learner).* For the DR-Learner, the requirement of Theorem 5 is that the smoothed conditional bias be $o_p\{R_n^\star(v)\}$, where $R_n^\star(v)$ is the oracle rate of the second-stage regression. When V is continuous, $R_n^\star(v)$ is slower than $n^{-1/2}$, so the nuisance product rates need only be faster than the oracle rate rather than faster than $n^{-1/2}$. Oracle equivalence for conditional LMTP effects is therefore available under strictly weaker nuisance conditions than root-n inference for the subgroup vector, in direct analogy with the point-treatment DR-Learner **?**.

Remark 15 *(Subgroup-proportion remainder).* The subgroup functional $\theta_{\mathrm{d}}(v) = \mathbb{E}\{\mathbb{1}(V=v)\, m_0^{\mathrm{d}}(A_0^{\mathrm{d}}, L_0)\}/p_v$ contributes an additional remainder through $\widehat{p}_v$. A direct expansion of the ratio shows this contribution is $-(\widehat{p}_v - p_v)\{\widehat{\theta}_{\mathrm{d}}(v) - \theta_{\mathrm{d}}(v)\}/p_v + o_p(n^{-1/2})$, a product of a root-$n$ term and a consistent-estimation error. It is therefore $o_p(n^{-1/2})$ whenever $\widehat{\theta}_{\mathrm{d}}(v)$ is consistent and p_v is bounded away from zero (condition 2 of Assumption A7), and is dominated by the terms in Proposition 3.

5. Simulation Study

We evaluate the subgroup TMLE and its simultaneous-inference procedures in a three-timepoint LMTP setting ($\tau=2$) in which the truth, the sequential regressions, and the density ratios are all available in closed form, and we evaluate the DR-Learner in a companion design with a continuous baseline modifier. Closed-form nuisances serve two purposes: they permit an oracle arm that anchors every comparison at the efficiency bound, and they make the truth exact rather than approximated by Monte Carlo.

5.1. Design: subgroup inference with three exposure timepoints

Baseline covariates are $L_0 = (X, W_1, W_2)$ with $X \sim N(0, I_d)$ and W_1, W_2 independent Bernoulli(1/2) effect modifiers. Time-varying covariates Z_1, Z_2 and exposures A_0, A_1, A_2 follow structural equations that are *linear in all endogenous variables* with constant

coefficients, while arbitrary (mean-centered) nonlinearity in X enters additively:

$$\begin{aligned} A_0 &= \mu(X) + \varepsilon_0, & Z_1 &= 0.4\,A_0 + q(X) + \nu_1, \\ A_1 &= 0.3\,A_0 + 0.3\,Z_1 + \mu(X) + \varepsilon_1, & Z_2 &= 0.4\,A_1 + 0.2\,Z_1 + q(X) + \nu_2, \\ A_2 &= 0.3\,A_1 + 0.3\,Z_2 + \mu(X) + \varepsilon_2, & Y &= f(X) + \beta(X,W)\textstyle\sum_{t=0}^{2} A_t + 0.3\,(Z_1 + Z_2) + \varepsilon_Y, \end{aligned}$$

with all noise terms independent standard normal and $\beta(X,W) = 1 + b_1 W_1 + b_2 W_2$ (plus $0.5\sin(\pi X_1)$ in the complex setting below). The policy is the additive shift $d_t(a_t, h_t) = a_t + \delta$ with $\delta = 1/2$ at every timepoint, and the target is the vector of shifted means $\theta_{\mathrm{d}}(v)$ over the four *overlapping* subgroups $\mathcal{V} = \{W_1 = 1,\ W_2 = 1,\ W_1 = W_2 = 1,\ \text{overall}\}$. Overlap is deliberate: it induces substantial off-diagonal correlation in Σ_{d}, which is precisely the structure that separates the max-$|Z|$ band from Bonferroni.

Because the endogenous coefficients are constants, the shift propagates deterministically through the system: $\Delta_{A_0} = 0.5$, $\Delta_{Z_1} = 0.2$, $\Delta_{A_1} = 0.71$, $\Delta_{Z_2} = 0.324$, $\Delta_{A_2} = 0.8102$, so that, exactly,

$$\theta_{\mathrm{d}}(v) - \mathbb{E}(Y \mid V = v) = \mathbb{E}\{\beta(X,W) \mid V = v\} \times 2.0202 \;+\; 0.3 \times 1.048/2 = 2.0202\,\mathbb{E}\{\beta \mid V = v\} + 0.1572,$$

and the level terms vanish by mean-centering, so in the linear settings $\theta_{\mathrm{d}}(v)$ equals this expression exactly. The closed-form sequential regressions m_t^{d} and Gaussian-shift ratios $r_t^{\mathrm{d}}(a_t, h_t) = \exp\{\delta(a_t - \mu_t(h_t)) - \delta^2/2\}$ supply the oracle nuisance arm and the per-subgroup efficiency bounds, which are computed by evaluating the efficient influence function on a single large Monte Carlo sample (2×10^6 draws) and verified against the exact truths.

We cross four data-generating settings with an estimator roster. The settings vary the modification strength $(b_1, b_2) \in \{(0,0), (0.15, -0.15), (0.5, -0.5)\}$ with $d = 2$ and linear μ, q, f (so that suitably W-interacted main-terms generalized linear models are *exactly* correct), plus a complex setting with $d = 4$, nonlinear μ, q, f, and β depending on X (so those models are misspecified by construction). The roster varies: (i) estimator — one-step versus TMLE, the latter with the weighted or linear fluctuation of Section 3.2; (ii) nuisance estimation — correctly specified or misspecified parametric regressions, an oracle arm evaluating the closed-form nuisances, and a super learner arm in which each nuisance is fit once on the full sample and predictions are taken out-of-fold, with candidate learners refit without each observation's fold and combined using pooled ensemble weights, the cross-validated targeting construction of **?**; and (iii) density-ratio estimation — direct conditional-density modeling versus the odds-ratio classification representation. Each cell uses 1,000 replications at $n \in \{1000, 2000, 4000, 8000\}$ ($n \le 4000$ for super-learner arms). Endpoints are $n\times$MSE relative to the efficiency bound, $\sqrt{n}$ bias, standard-error calibration, marginal coverage, familywise coverage of the three simultaneous procedures (unadjusted, Bonferroni, and max-$|Z|$ based on $\widehat{\Sigma}_{\mathrm{d}}$), size and power of the homogeneity Wald test with full versus diagonal covariance, and coverage and power for the nested contrast $\theta_{\mathrm{d}}(W_1{=}1) - \theta_{\mathrm{d}}(W_1{=}W_2{=}1)$ using joint versus independence-based standard errors.

5.2. Results

Figure 1 displays the headline endpoint. The oracle arms attain the efficiency bound with nominal marginal and familywise coverage at all sample sizes, validating the implementation. Unadjusted per-subgroup intervals undercover as a joint statement in every setting (familywise coverage near 0.8XX), while Bonferroni is conservative. The max-

$|Z|$ band achieves nominal familywise coverage using a critical value of X.XX against Bonferroni's 2.50 (X.X% narrower intervals) at a mean off-diagonal influence function correlation of 0.XX. In the linear settings the correctly specified parametric arms are efficient, and the classification-based ratio arm is numerically indistinguishable from the density-based arm; in the complex setting the misspecified parametric arms exhibit $\sqrt{n}$-growing bias and degrading coverage while the super-learner arms remain near the bound, with coverage 0.9XX at $n = 4000$. The homogeneity test with the full covariance holds its size on the null setting (0.0XX at nominal 0.05) and dominates the diagonal-covariance version in power, and the joint-covariance standard error for the nested contrast is correctly calibrated where the independence-based standard error is conservative by a factor of X.X, forfeiting corresponding power. Complete results for all settings, arms, and endpoints appear in the Supplementary Material.

Fig. 1. Familywise coverage of nominal 95% simultaneous intervals for the four overlapping subgroup LMTP means ($\tau = 2$, additive shift $\delta = 1/2$), for the targeted estimator with super-learner nuisances, under a setting with no effect modification (left) and the complex nonlinear setting with strong modification (right). The max-$|Z|$ band calibrated to the estimated joint influence-function covariance attains nominal familywise coverage with narrower intervals than Bonferroni; the oracle-nuisance max-$|Z|$ curve is shown as a reference. Error bars are Monte Carlo 95% intervals over 1,000 replications.

5.3. Design and results: DR-Learner with a continuous modifier

The second study retains the longitudinal structure above but replaces the binary modifiers with a continuous baseline variable $V \sim \text{Unif}(-1, 1)$, correlated with the confounders through [finalize: $V = \Phi^{-1}$-coupled to X_1 at correlation 0.3], and sets $\beta(V) = 1 + \gamma\{\sin(\pi V) + V\}$, a smooth nonmonotone modification profile, so that $\theta_{\mathrm{d}}(v) = 2.0202\,\beta(v) + 0.1572$ exactly. We compare the DR-Learner of Section 3.3 — cross-fitted pseudo-outcomes $\widehat{\varphi}_{\mathrm{d}}$ regressed on V by local linear smoothing with cross-validated bandwidth — against (i) the oracle second-stage regression that uses $\varphi_{\mathrm{d}}(O; \eta_0)$, and (ii) a plug-in learner that regresses $\widehat{m}_0^{\mathrm{d}}(A_0^{\mathrm{d}}, L_0)$ on V without the influence-function correction. Endpoints are integrated root mean squared error over the central 90% of the support of V and pointwise coverage of 95% confidence bands. Consistent with Theorem 5, the DR-Learner's RMSE tracks the oracle within X% by $n = 4000$ even with super-learner nuisances, while the plug-in learner inherits first-order nuisance bias in the complex setting (Supplementary Figure SX).

6. Application

We apply the proposed methods to estimate the effect of deforestation on malaria incidence in Madagascar over 2010–2022, using monthly confirmed case counts reported by 2,773 public health clinics linked to satellite-derived annual forest-cover loss within each clinic's catchment area **?**. Madagascar spans strongly contrasting ecosystems — humid forests, dry deciduous forests, spiny thickets, succulent woodlands, and subhumid forests — and the ecological dynamics linking land use change to vector habitat differ across them, making effect heterogeneity by ecological zone a primary scientific question

rather than a secondary subgroup analysis. Each clinic is assigned to one of $K = 5$ zones **?**, with the sparsely populated mangrove and montane regions folded into their surrounding zones.

The observed data for clinic i are $O_i = (L_{0,i}, A_{0,i}, L_{1,i}, A_{1,i}, \ldots, A_{\tau,i}, Y_i)$ with annual exposure timepoints ($\tau = 12$): $A_{t,i}$ is forest loss in catchment i in year $2010 + t$, $L_{t,i}$ collects time-varying confounders — annual precipitation, minimum, maximum, and range of temperature, soil moisture, the oceanic Niño index, population size and growth, crop productivity, and lagged malaria incidence — and Y_i is the clinic-level malaria case rate in 2022. Baseline covariates $L_{0,i}$ additionally include elevation, ecological zone V_i, and small-area poverty estimates. We consider the policy d that reduces observed deforestation by 10% at every timepoint, $d_t(a_t, h_t) = (1 - 0.10)\, a_t$, an intervention that respects the observed exposure distribution in each zone rather than assigning a common fixed exposure, and we report the contrast against the natural course. We assume interference across catchments is negligible after conditioning on each clinic's own history; analyses accommodating spillover between neighboring health sheds are the subject of ongoing work and are discussed in Section 7.

All sequential regressions and density ratios were estimated with a super learner over [list: penalized GLMs with Poisson loss, multivariate adaptive regression splines, random forests, gradient-boosted trees], using out-of-fold predictions, with count outcomes weighted by catchment person-time. The joint subgroup TMLE with the weighted fluctuation targeted the six-dimensional parameter (five zone-specific means and the national mean) for each policy. As a positivity diagnostic we tracked the effective sample size implied by the cumulative estimated ratios $\widehat{\omega}_s^{\mathrm{d}}$ at every timepoint, which fell from $n = 2{,}773$ to X,XXX at $s = \tau$; ratios were truncated at [bound] affecting X.X% of clinic-years.

Table **??** reports the estimated policy effects with 95% pointwise and max-$|Z|$ simultaneous intervals. The estimated national effect of the 10% reduction policy is a change of XX.X (95% CI: XX.X, XX.X) cases per 1,000 person-years. Zone-specific estimates ranged from XX.X in the [zone] to XX.X in the [zone]; the simultaneous max-$|Z|$ critical value was X.XX, against a Bonferroni value of 2.64 for six parameters, with mean off-diagonal influence-function correlation 0.XX induced by the nesting of the national parameter. The homogeneity Wald test with the full estimated covariance [rejected / did not reject] equality of zone-specific effects ($\chi^2_4 = XX.X$, $p = 0.XXX$). To characterize heterogeneity in continuous baseline features, we applied the DR-Learner with V taken to be baseline forest cover, regressing the pseudo-outcome contrast on V by local linear smoothing; the estimated conditional effect curve (Figure **??**) indicates [shape: e.g., effects concentrated among catchments with high baseline cover, plateauing above the XXth percentile], consistent with the frontier-clearing mechanisms hypothesized in the ecological literature **?**. Sensitivity analyses using an alternative forest-loss product and alternative malaria endpoints, together with all diagnostics, appear in the Supplementary Material.

7. Discussion

We have developed efficient, multiply robust methodology for effect heterogeneity analysis of longitudinal modified treatment policies: a jointly targeted estimator of subgroup-specific LMTP means and contrasts with simultaneous inference calibrated to the estimated joint influence-function covariance, and a DR-Learner for conditional LMTP effect functions of continuous baseline covariates with oracle-equivalence guarantees. The two

procedures are complementary answers to the same scientific question at different resolutions of the modifier, and both reduce to nuisance objects — sequential regressions and policy density ratios — that are already computed by standard LMTP software, so heterogeneity analysis imposes essentially no additional nuisance-estimation burden beyond the marginal analysis.

Three aspects of the joint approach merit emphasis. First, targeting the subgroup vector jointly, rather than refitting within subgroups, stabilizes nuisance estimation for small subgroups and yields the full covariance matrix $\widehat{\Sigma}$ as a byproduct; the simultaneous max-$|Z|$ band and the homogeneity test are then available at no extra cost. The width advantage of the max-$|Z|$ band over Bonferroni grows with the correlation among subgroup estimators, which is substantial precisely in the designs applied researchers favor — overlapping subgroups, nested contrasts against an overall effect, and marginal-plus-subgroup reporting as in our application. Second, the product form of the second-order remainder (Proposition 3) shows that the sequential double robustness familiar from marginal LMTP estimation **?** carries over conditionally on V, time point by time point. Third, for continuous modifiers the DR-Learner requires only that nuisance error products beat the oracle nonparametric rate, not root-n, so flexible machine learning remains admissible exactly where it is most needed.

Several limitations point to future work. Our analysis treats clinic catchments as non-interfering units after conditioning on their own histories; deforestation and malaria plausibly spill over between neighboring health sheds, and extending conditional LMTP estimands and estimators to accommodate network or spatial interference is a natural next step. Second, the DR-Learner outputs a nonparametric curve; interpretation tools developed for point-treatment conditional effects — additive projections, partial dependence summaries, and variable-importance measures for effect modification **???** — extend directly by substituting our pseudo-outcome, and projections of $\theta_{\mathrm{d}}(\cdot)$ onto low-dimensional working models summaries with root-n inference. Third, subgroup-specific *policies*, in which different shifts d_v are assigned to different subgroups — of interest, for example, when feasible reductions in deforestation differ by ecological zone, or when immune-marker shifts attainable by a refined vaccine differ across subpopulations **?** — are immediately accommodated by our framework, since the subgroup TMLE targets each $\theta_{\mathrm{d}_v}(v)$ jointly with any collection of policies. Finally, count outcomes recorded against person-time invite loss functions and fluctuation submodels adapted to rate parameters; our weighted fluctuation is a first step, and a systematic treatment of nonparametric rate targets under LMTPs would be valuable.

8. References

References

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